

Introducing the BKI-Ludwig Pan-Cancer T Cell Atlas

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On behalf of Drs. Chi Dang, Zhen Zeng, Drew Pardoll, Hongkai Ji,
Weidong Cao and the entire pan cancer team



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What problems are we trying to solve?

- ❑ Existing single cell transcriptomic datasets are fragmented
- ❑ Current pan-cancer studies are incomplete
- ❑ Common vs. unique pan-cancer T cell states and cancer-specific T cell features remain poorly defined
- ❑ Correlation between clinical metadata and T cell state is lacking, but is essential for translational discovery

What are we proposing to do?

Build the most comprehensive, open-access pan-cancer T cell resource, integrating single T cell transcriptomic data with extensive clinical metadata and innovative analytical tools into a freely available web app supported by a novel **user friendly** AI agent to accelerate immunotherapy discovery.

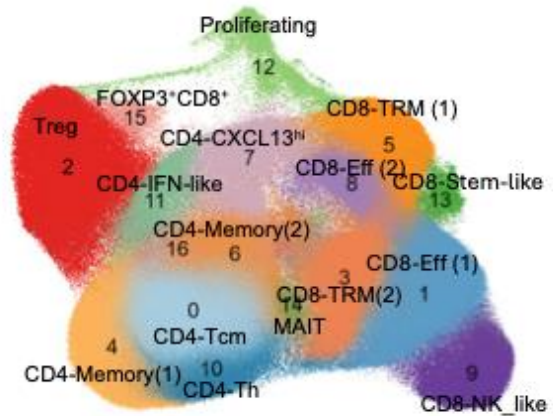
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How will our atlas be different?

Existing Approach: Reclustering

Re-integrate, re-cluster, and re-annotate
every time new data are added



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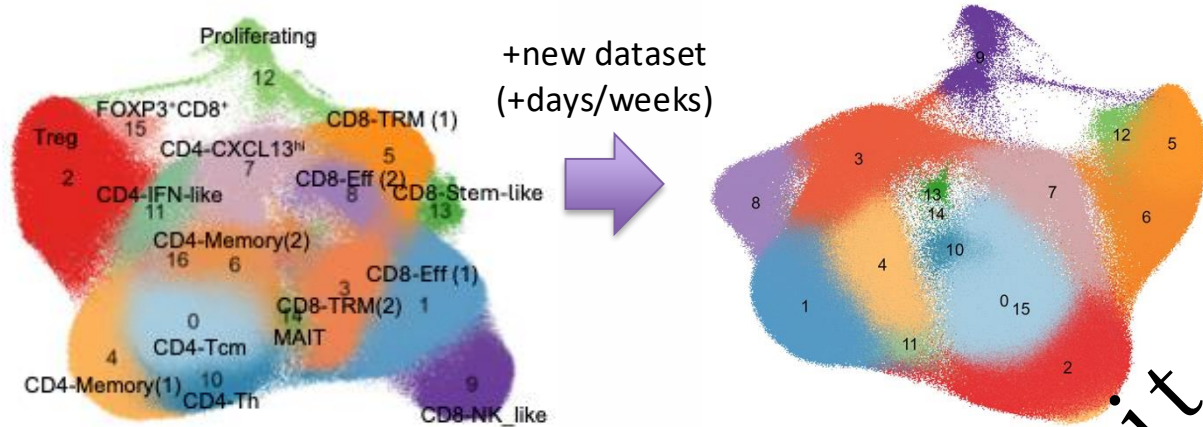
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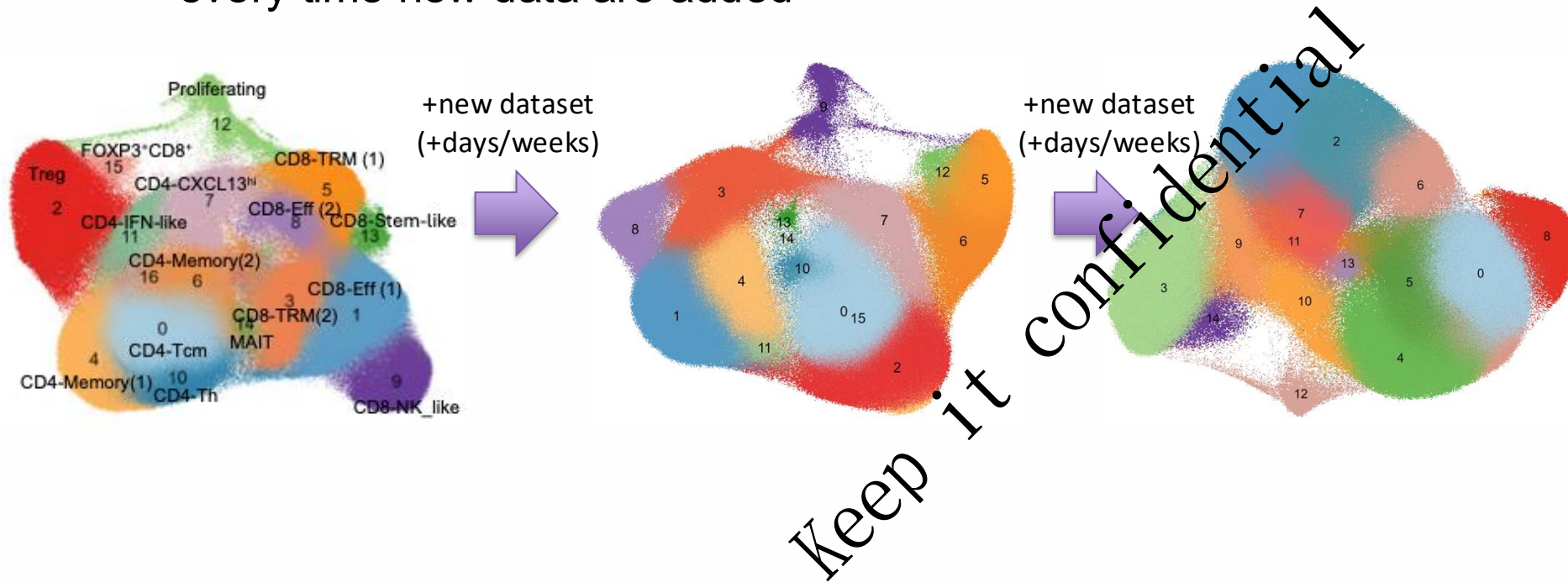
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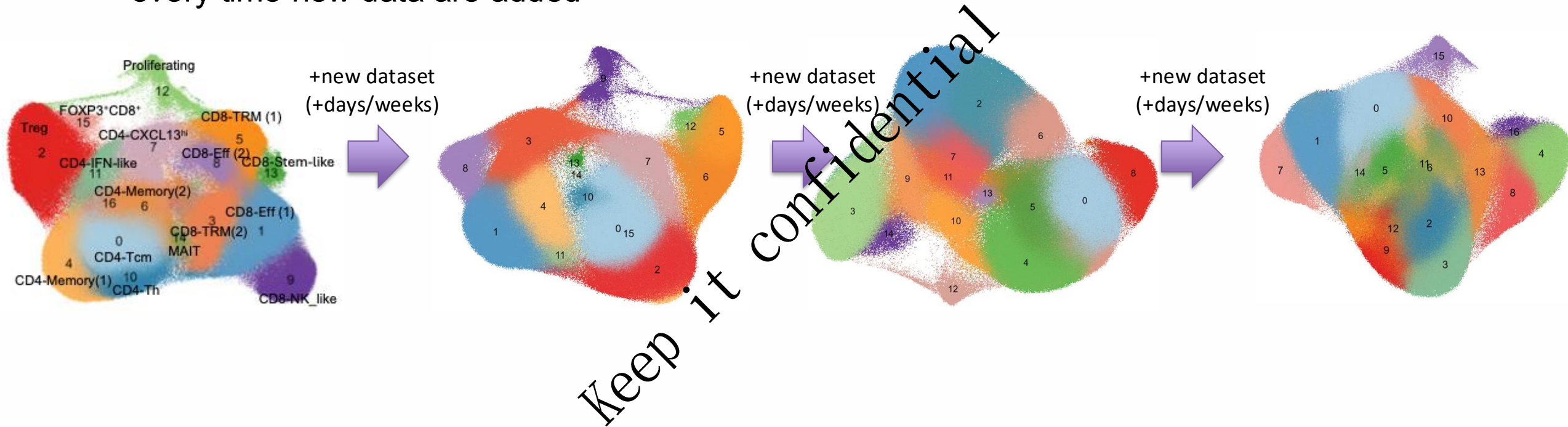
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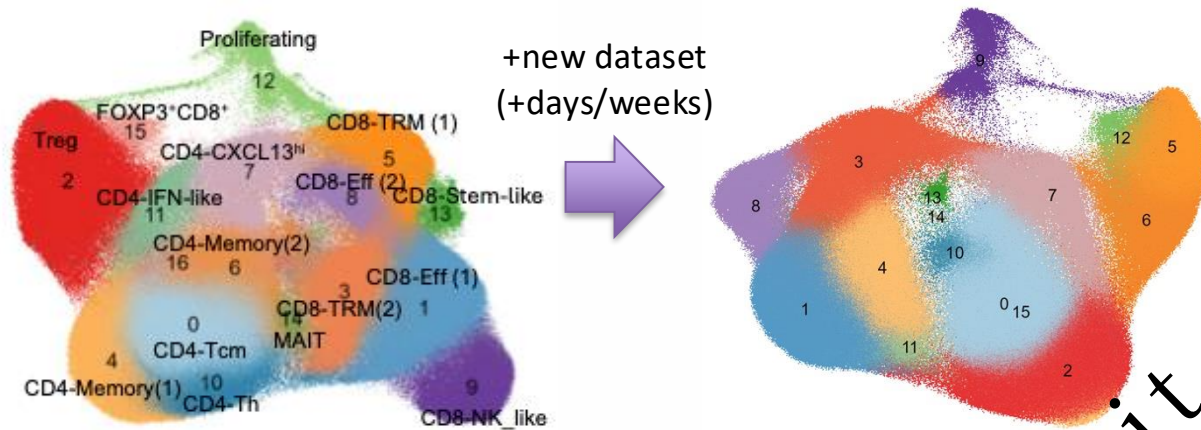
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SLOW Scales poorly, compute grows with every new dataset

UNSTABLE Clusters and annotations shift with each addition

INCONSISTENT Cell state labels constantly changing

MANUAL Requires re-annotation after every re-cluster



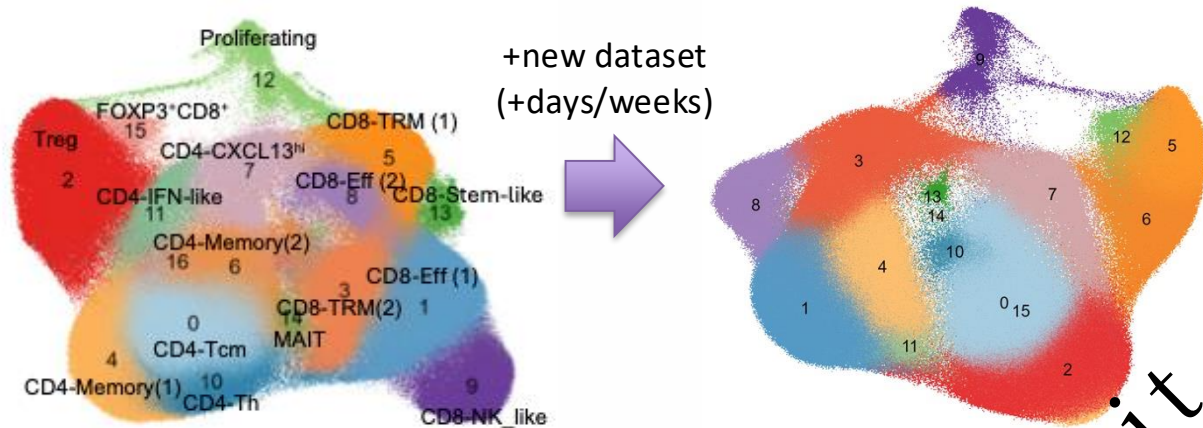
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Create the atlas once, project new data onto it

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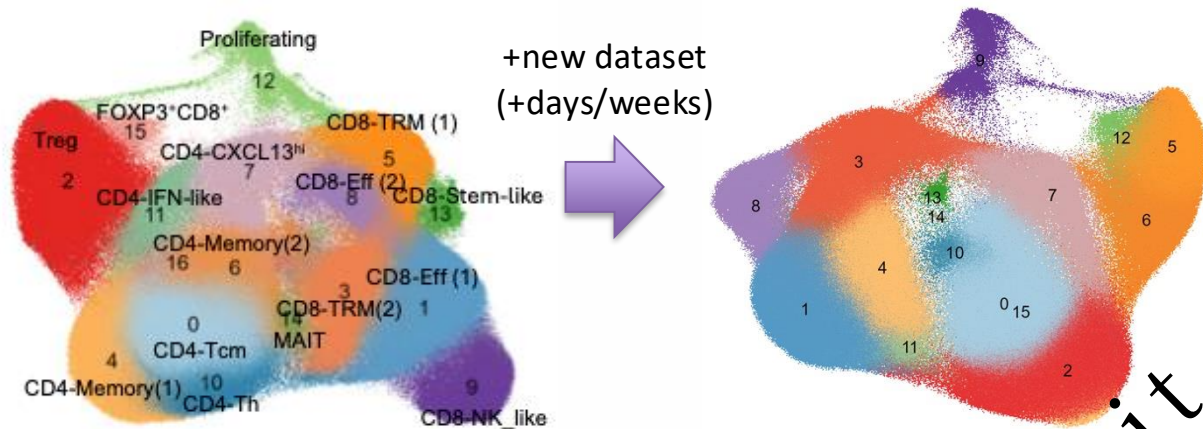
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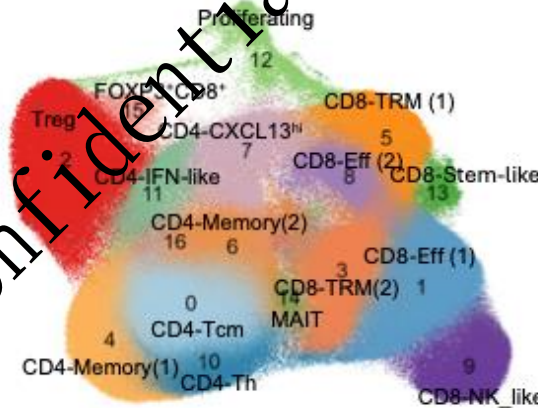
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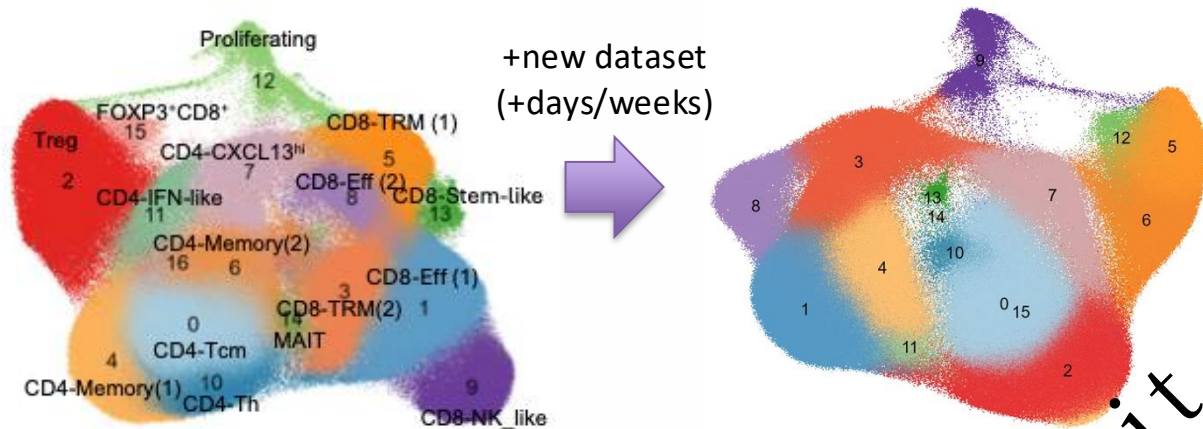
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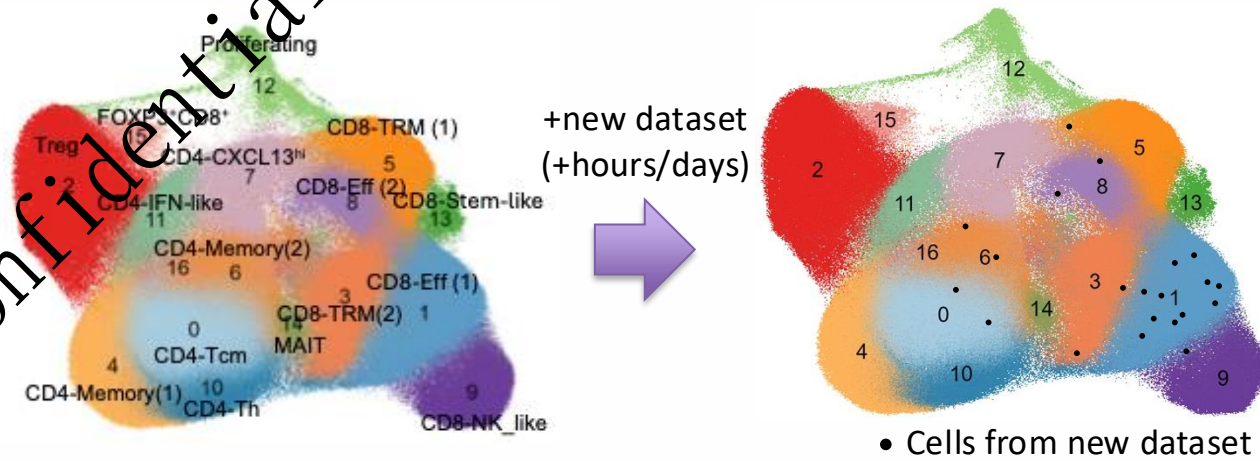


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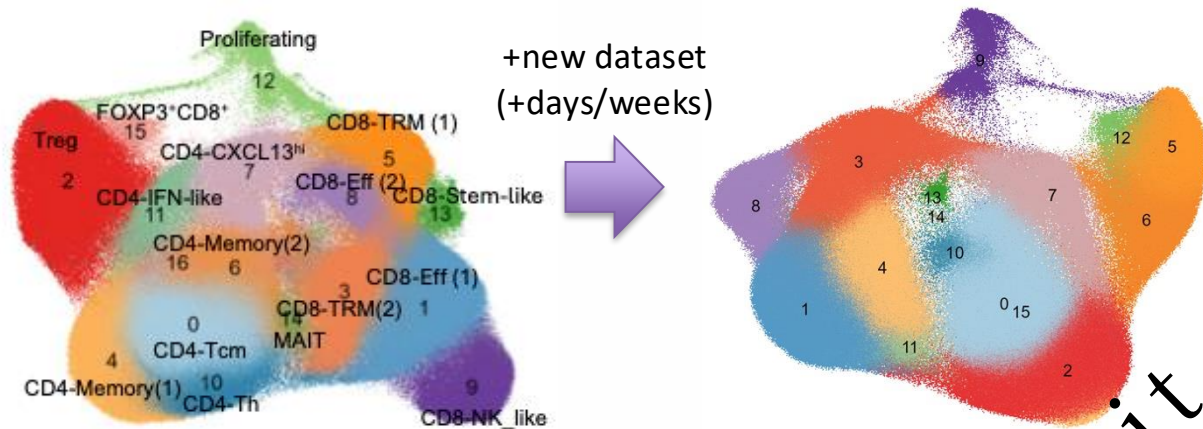


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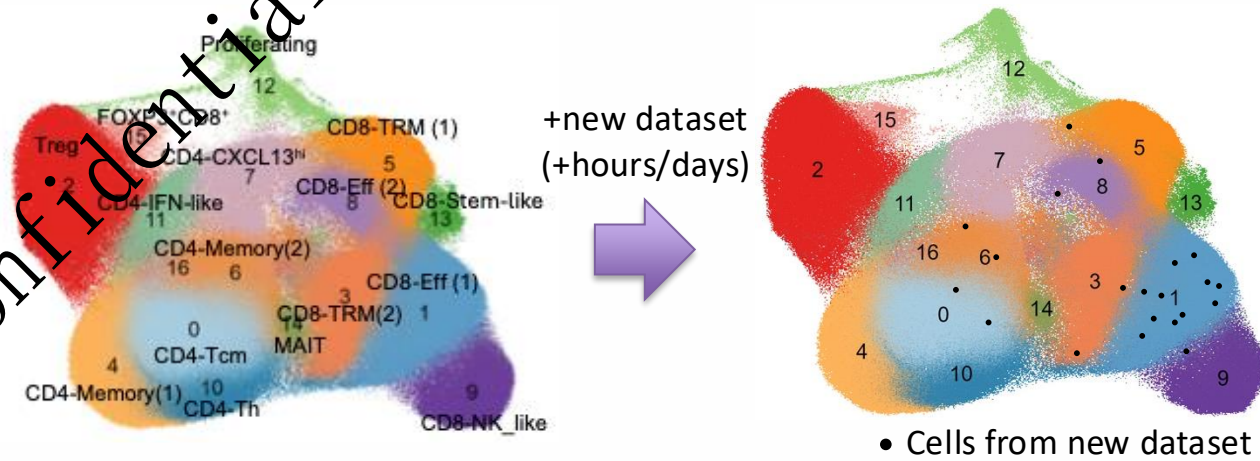
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- UNSTABLE** Clusters and annotations shift with each addition
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Our Approach: Reference Based Mapping

Create the atlas once, project new data onto it



- FAST** Only new cells are processed, not the entire dataset
- STABLE** Atlas coordinates and clusters never change
- CONSISTENT** Same cell state annotations regardless of new additions
- AUTOMATED** Label transfer without manual re-annotation



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How will our atlas be different?

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Reference-based mapping lets
new datasets integrate
without reclustering

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Extensive Metadata Curation

Meticulous annotation of clinical
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Batch & Patient Effect Correction

Removes technical and donor-specific confounders to isolate true biology

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Uniform QC Pipeline

Raw data processed identically across all datasets

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Explore & analyze data without computational expertise using integrated analytical platforms

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Dynamic, Living, Single T Cell Atlas

Discovery of pan-cancer and cancer type-specific T cell-based biomarkers, therapeutic targets, and biology



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Why is this important?

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Pan-cancer T cell assessments provide desperately needed context

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Example:

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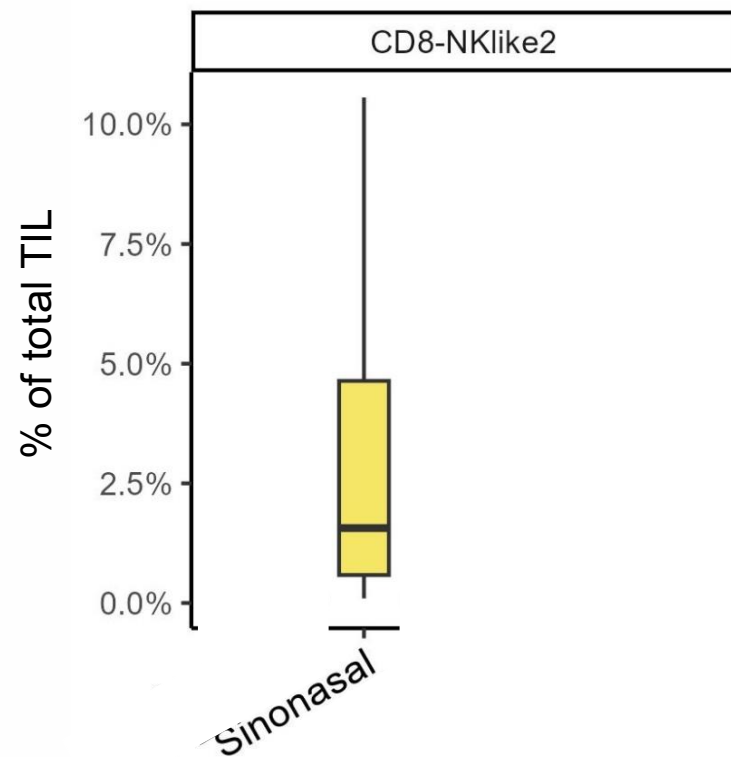
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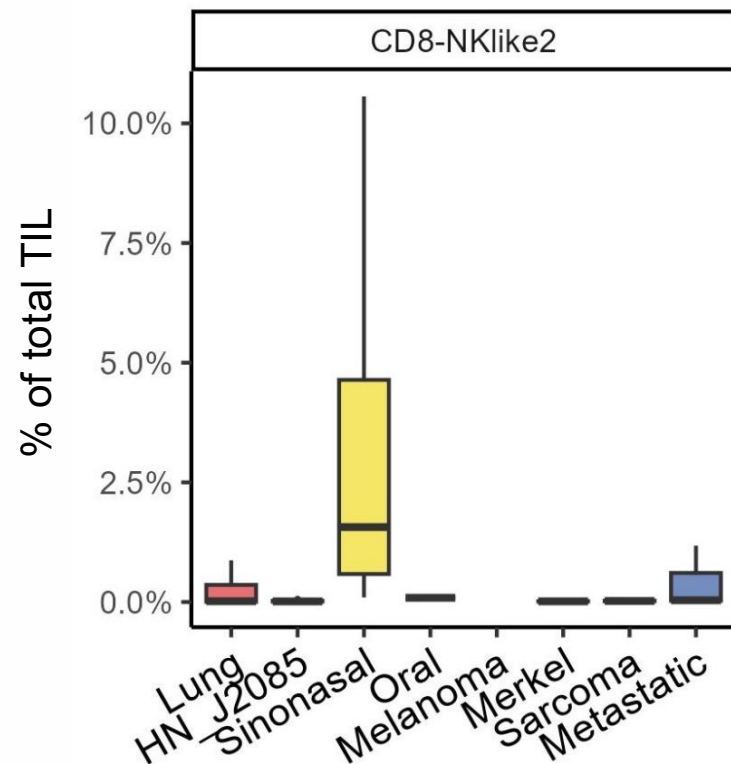
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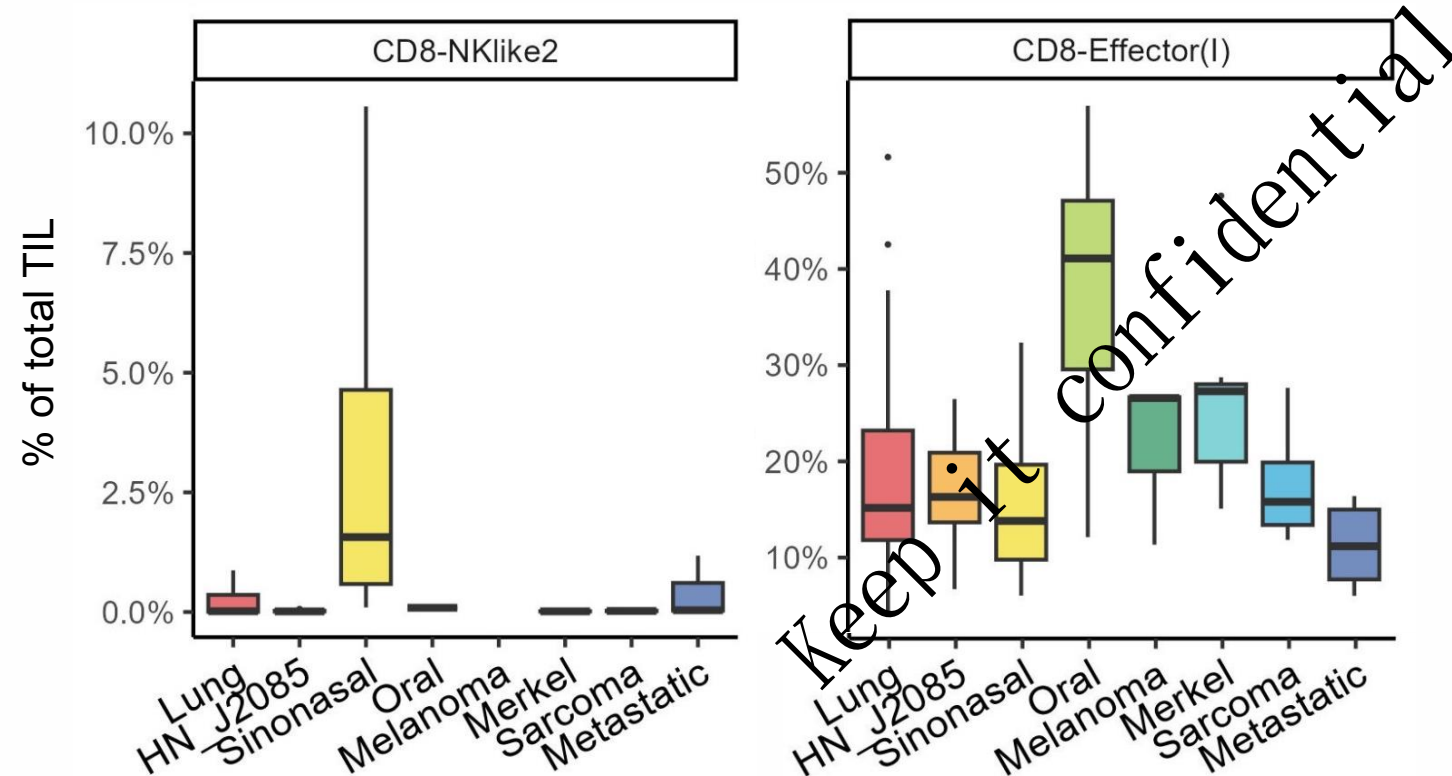
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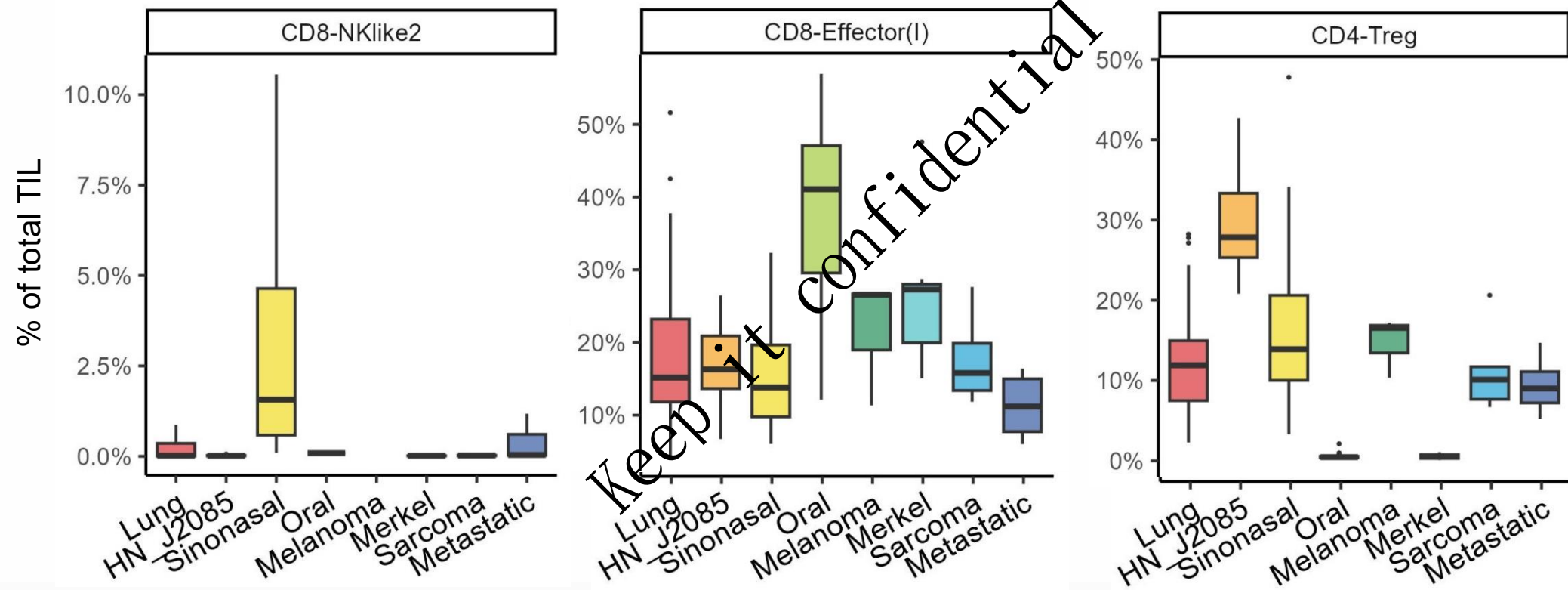
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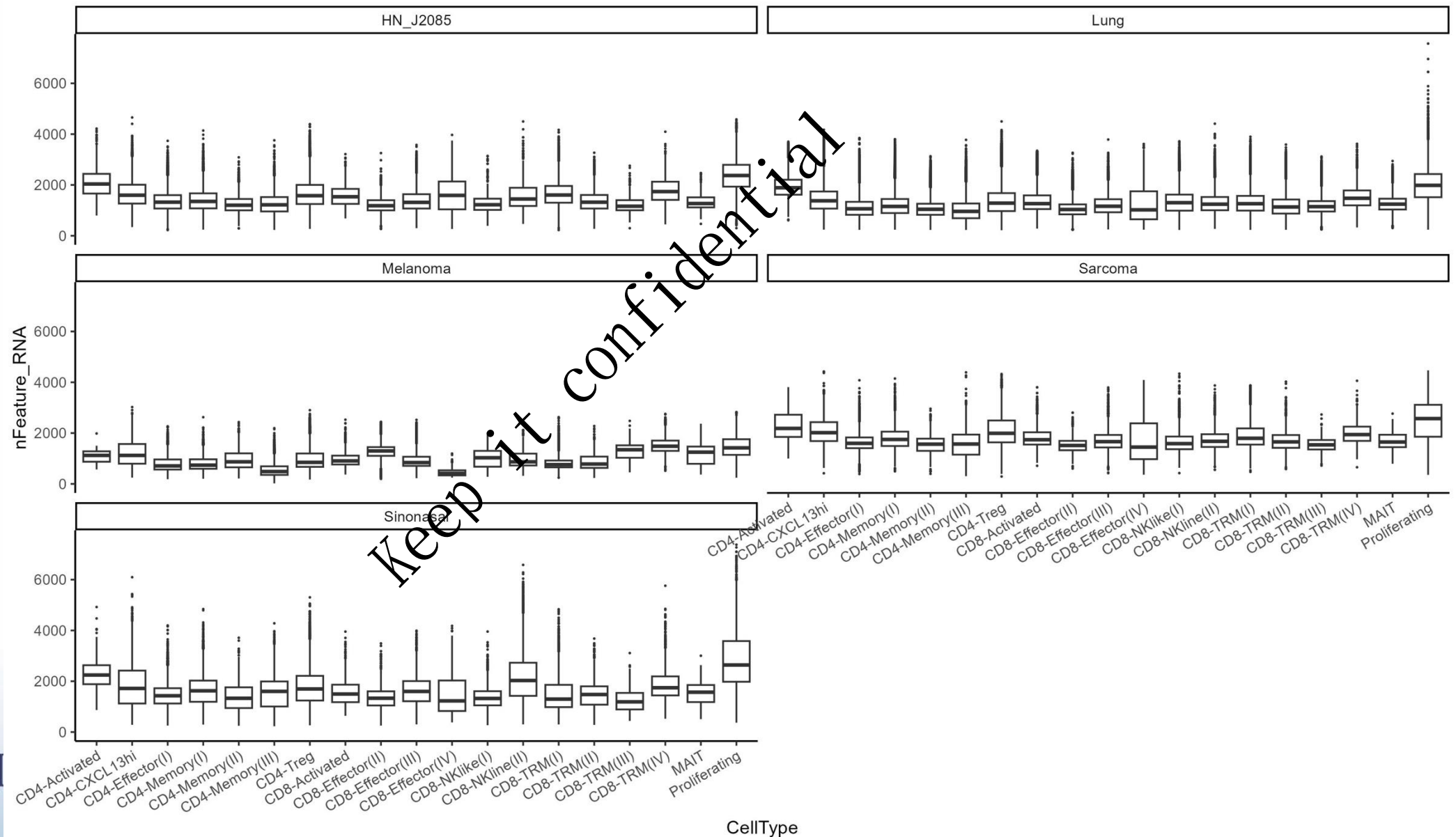
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Why is this important?

Pan-cancer T cell assessments can facilitate algorithm development

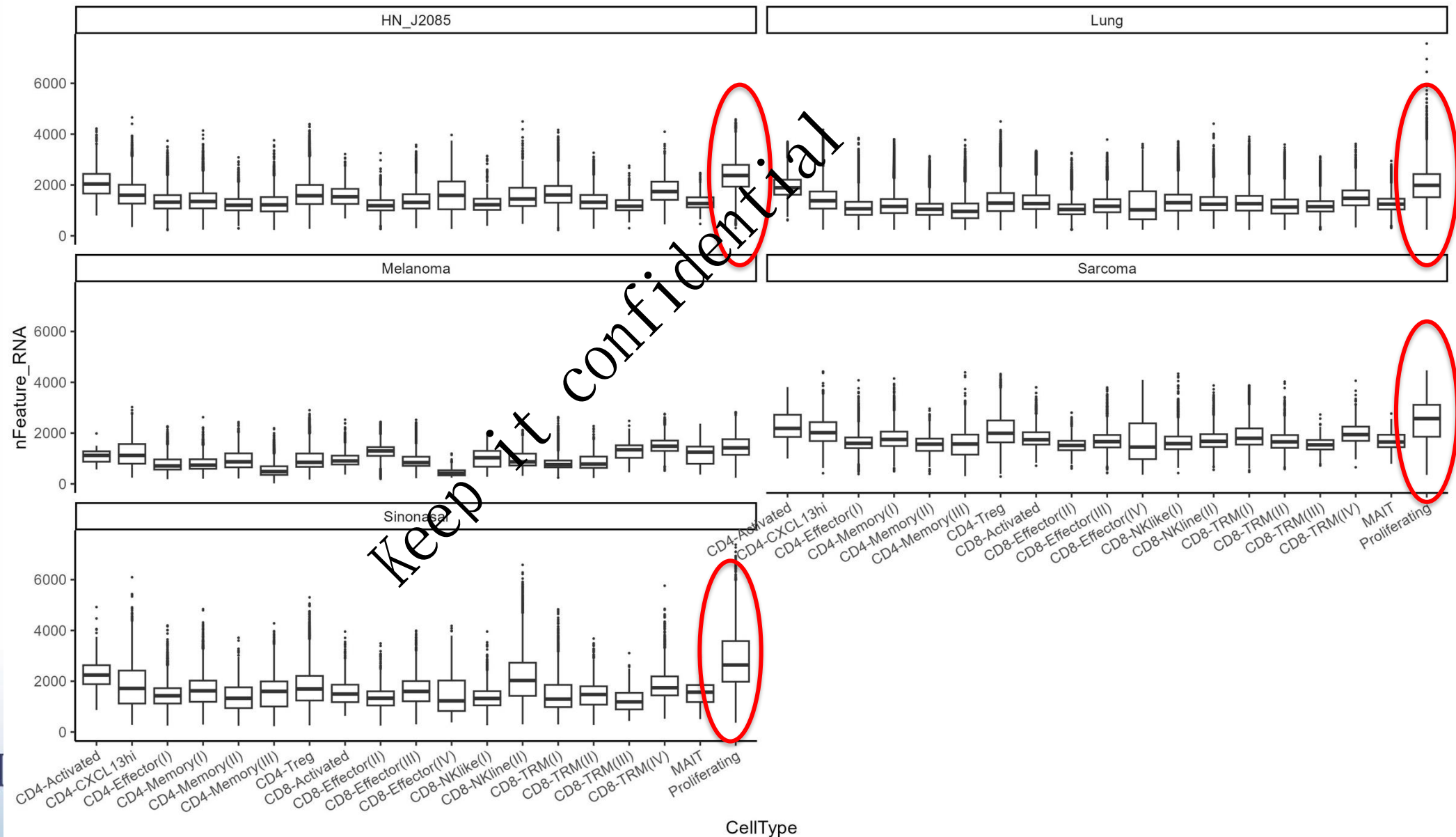
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Pan-cancer T cell assessments can facilitate algorithm development

Example:



What deliverables can be expected?

#1: Pan-cancer T cell atlas

20+

Cancer Types

3M+

T Cells

600+

Patients

1000+

Samples

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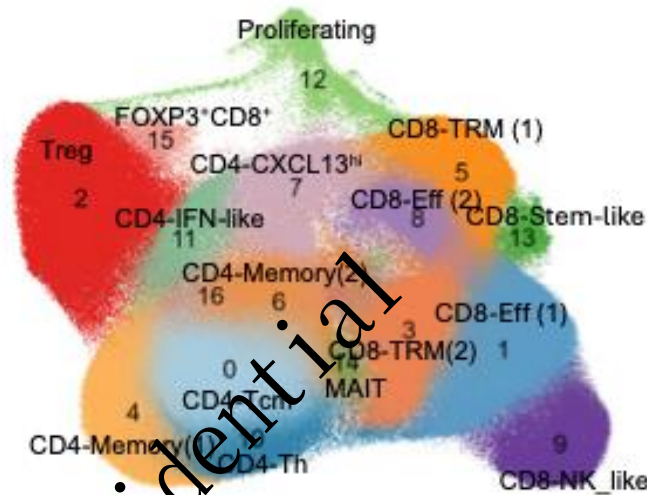
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CD3+ T Cell Reference Atlas



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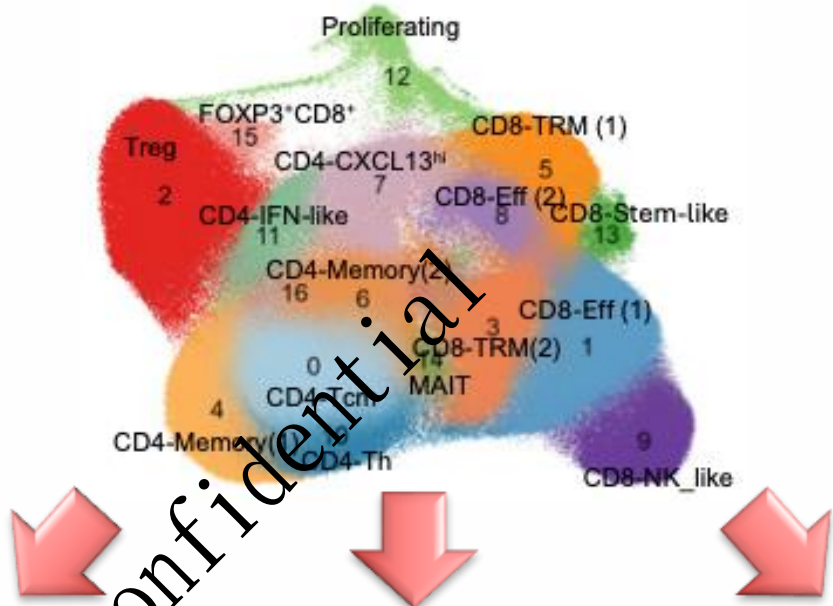
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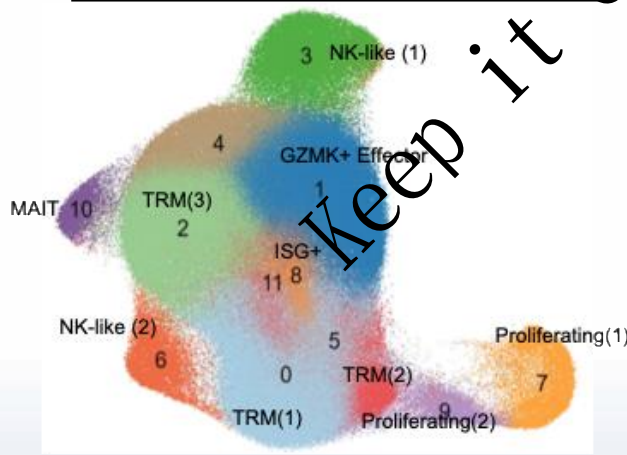
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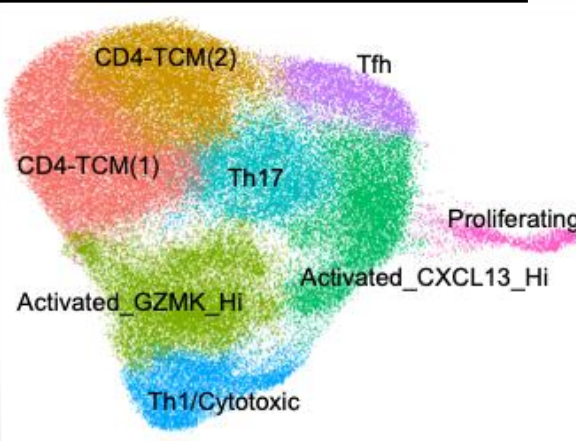
CD3+ T Cell Reference Atlas



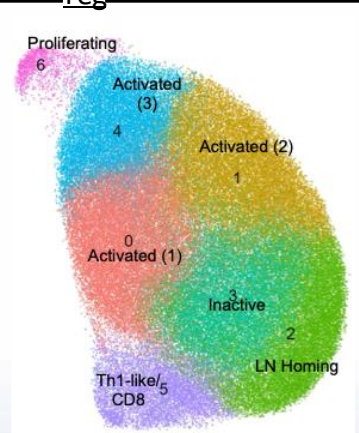
CD8+ T Cell Reference Atlas



CD4+ T Cell Reference Atlas



FoxP3+ T_{reg} Reference Atlas



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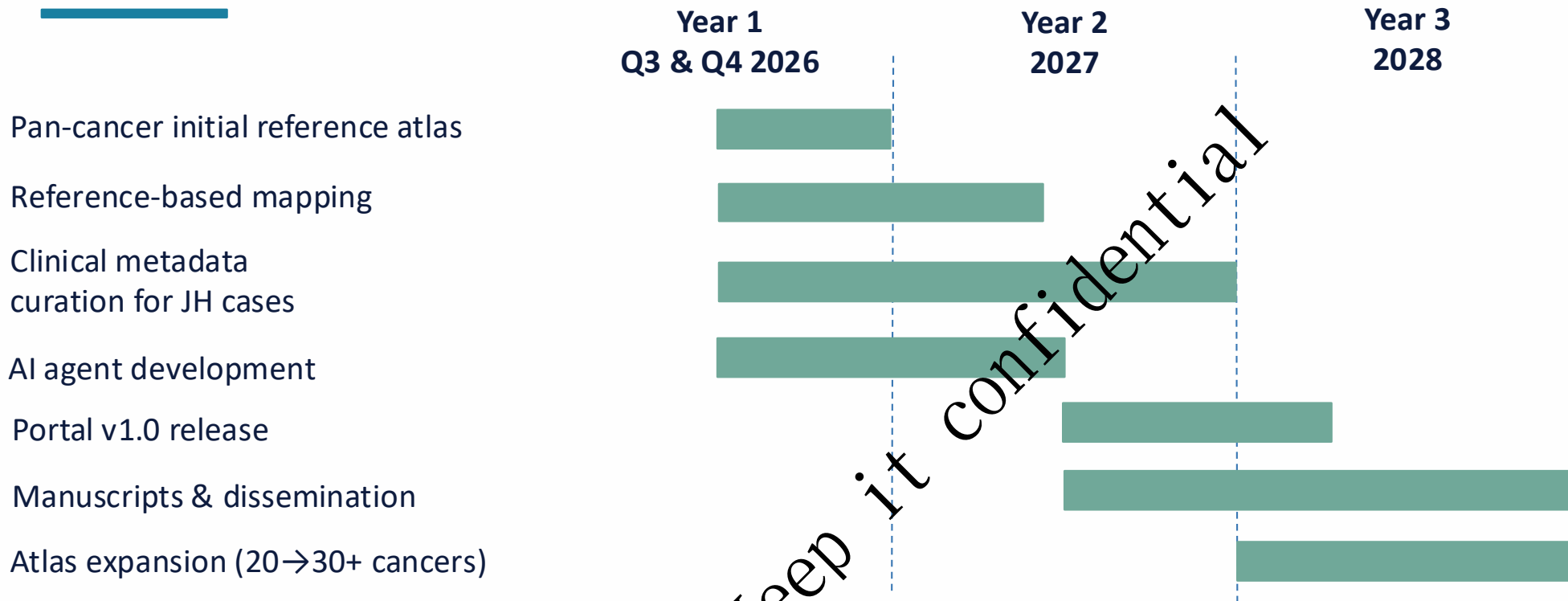
#2: Web app & AI interface

- ❑ A FREE community-facing platform enabling researchers worldwide to explore the atlas without computational barriers
- ❑ Integration of state-of-the-art computational algorithms and analytical pipelines
 - Published gene sets and scores, identification of tumor-reactive T cells, clonal trajectory pipelines, etc.
- ❑ Integration with a novel AI research agent
 - Natural language interface to query the atlas, ask questions about T cell states, tumor recognition, signature expression, and clonotype dynamics across cancer types without writing code

PLACEHOLDER FOR AI AGENT
FIGURE OR VIDEO FROM
HONGKAI AND TEAM



Timeline & Resource Allocation



Vision: 3+ years and beyond

Cancer type expansion

Add 10+ additional cancer types including rare and pediatric malignancies

Multimodal integration

Incorporate spatial transcriptomics to build a multi-omic T cell reference

Expanded clinical data from outside institutions

Pre-existing conditions, cancer genetics, HLA typing, lifestyle, etc

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Community resource analogous to TCGA for T cell biology across tumor types

Immunotherapy response prediction, biomarker discovery, drug target identification, cross-cancer academic discoveries, cell therapy design



THANK YOU!

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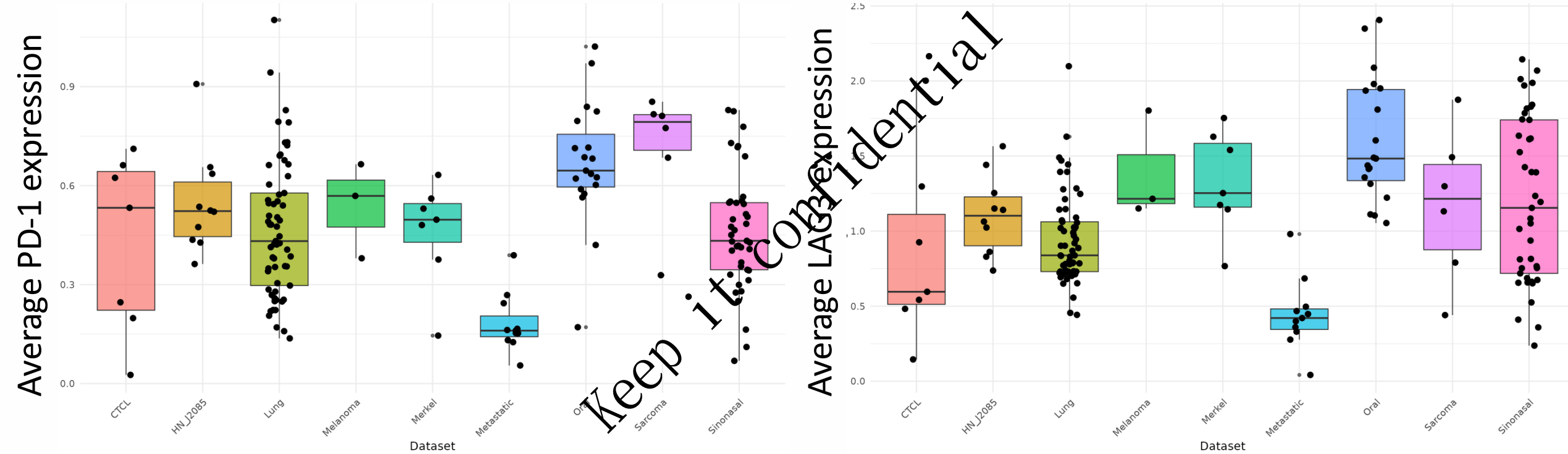


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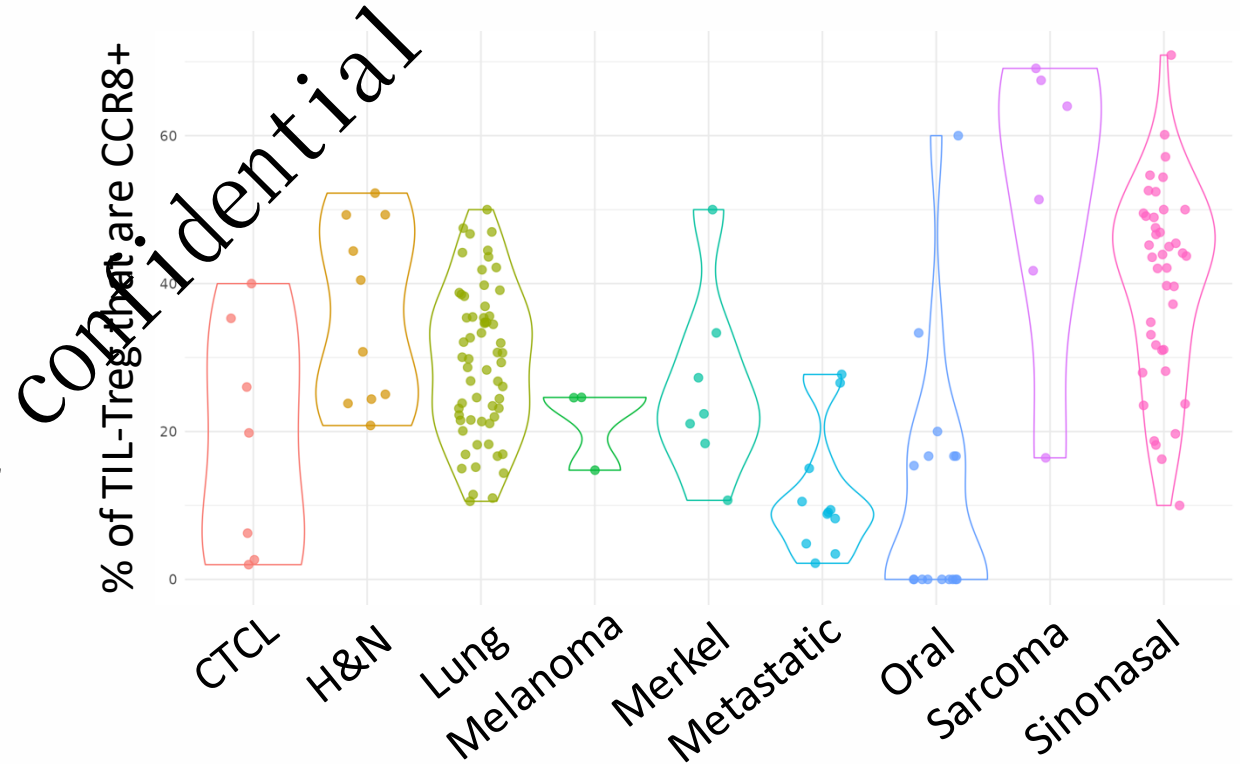
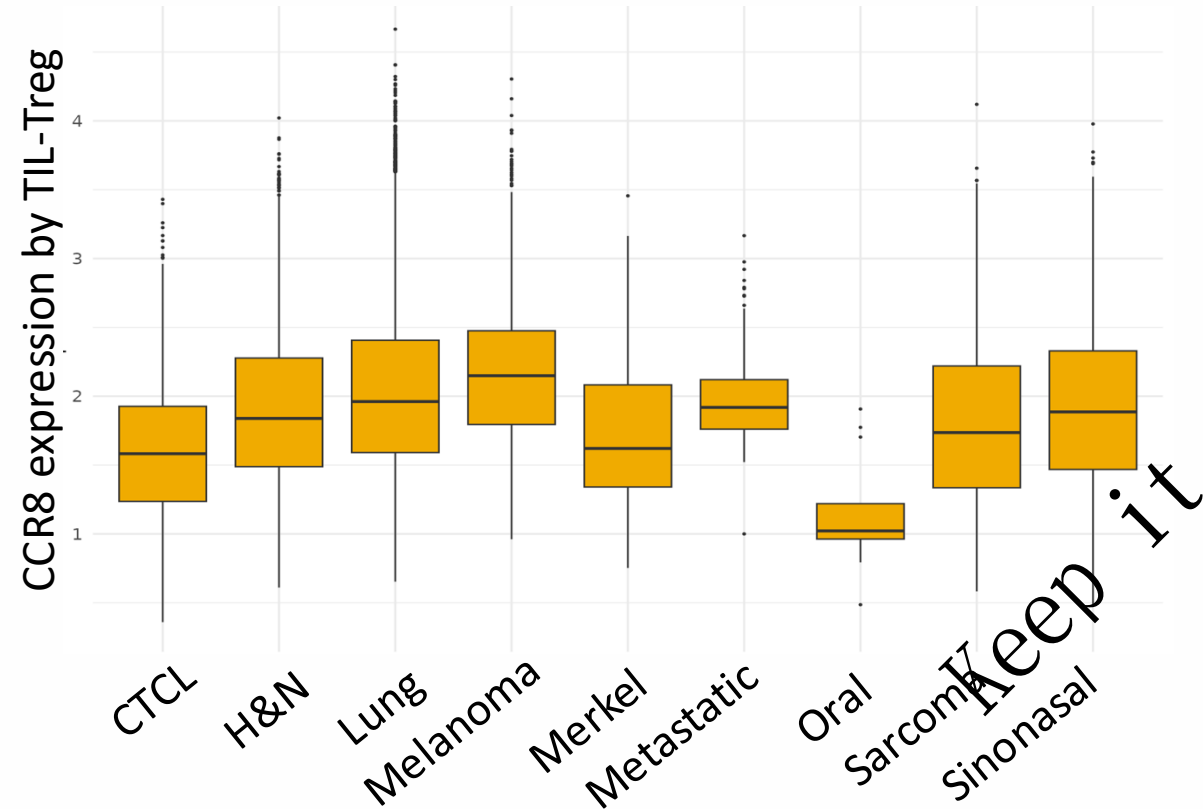


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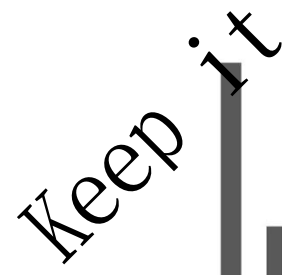
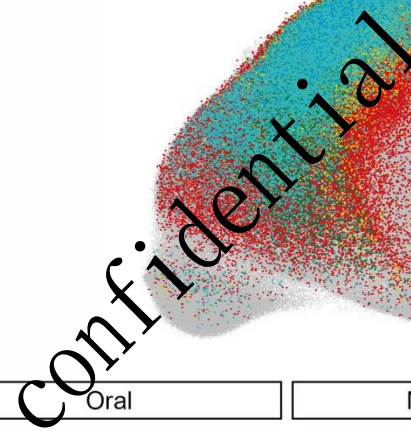




Pan cancer T cell assessments provide desperately needed context



pTRC localize to TRM across tumor types



Pan-Cancer T Cell Atlas Portal

MANAScore

ClonoTrace

TCR annotation

Metabolic State

Gene
expression

Survival

Custom
download

Dataset
submission

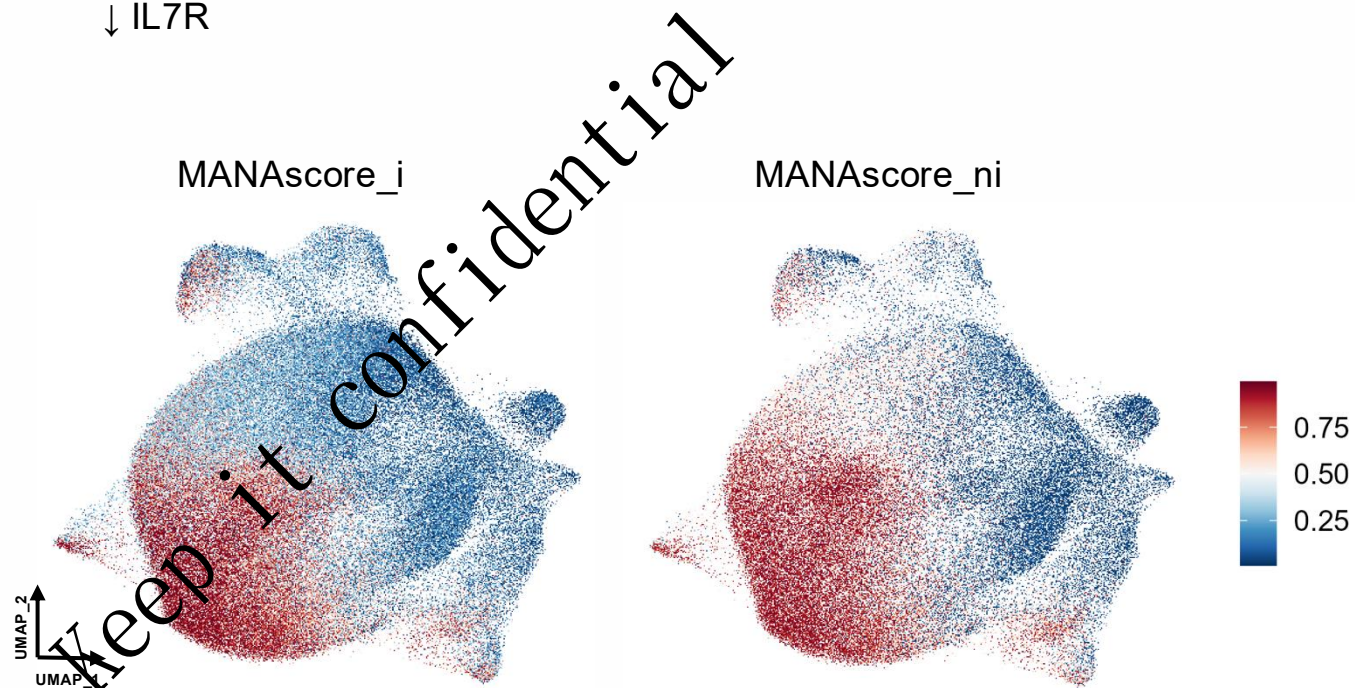
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MANAScore ~ CXCL13 + ENTPD1 – IL7R

↑ CXCL13

↑ ENTPD1 (CD39)

↓ IL7R



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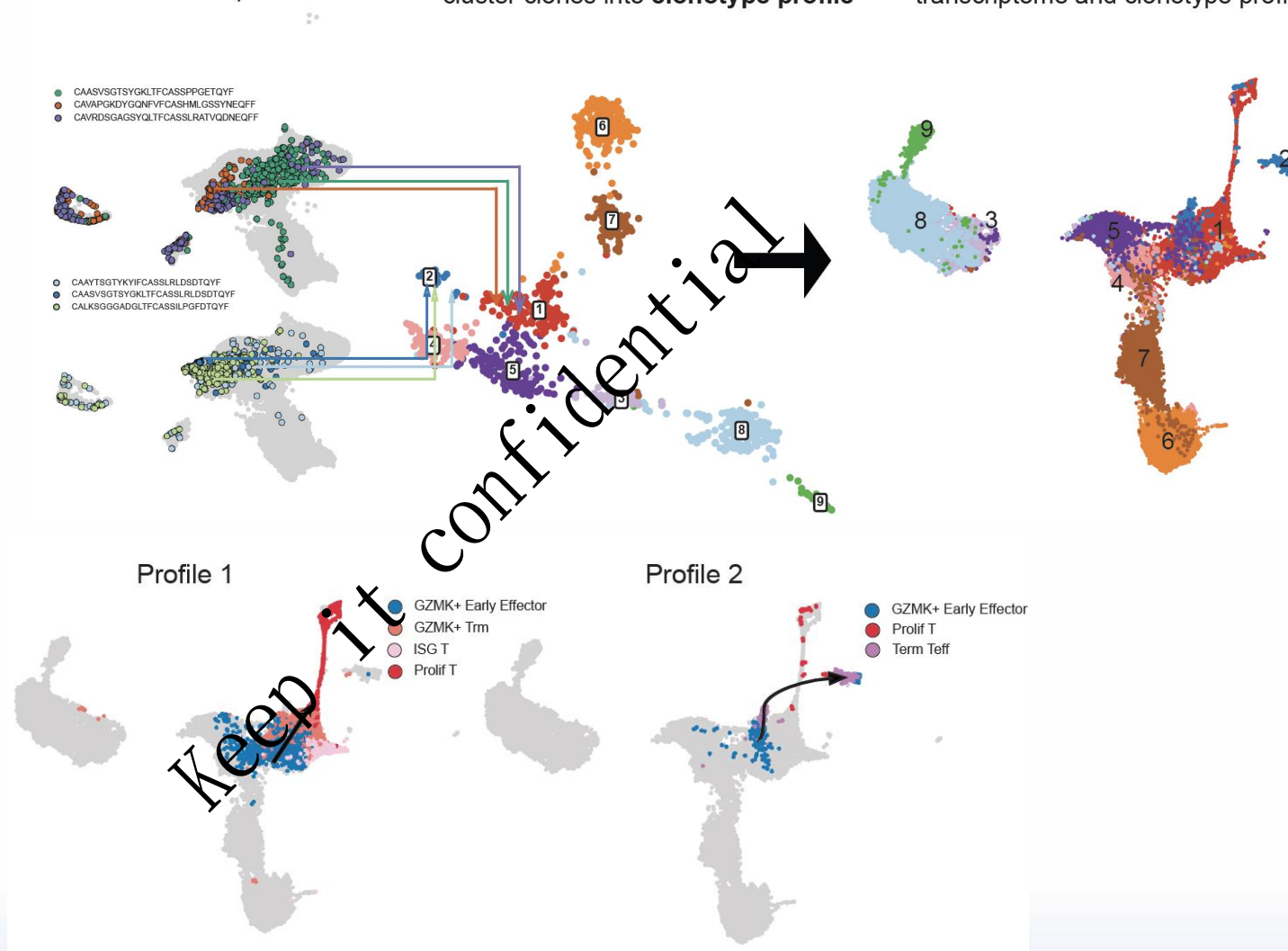
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... ..

Clonal development following
similar footprint

Using clonal footprint to embed clones and
cluster clones into **clonotype profile**

Refined UMAP integrating
transcriptome and clonotype profile



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Deliverable 3: Publication & website launch

